

Measures of Dispersion (Detailed)

Measures of Dispersion (also called **Measures of Variability** or **Measures of Spread**) are statistical measures that describe **how spread out or scattered the data values are around a central value (mean, median, or mode)**.

While **Measures of Central Tendency** tell us the center of the data, **Measures of Dispersion** tell us **how much the data varies**.

Definition

Measures of Dispersion are statistical measures that indicate the degree to which data values differ from each other and from the central value.

Why are Measures of Dispersion Important?

Imagine two classes with the same average (mean) score.

CLASS A

45 48 50 52 55

Mean = **50**

CLASS B

10 30 50 70 90

Mean = **50**

Both classes have the **same mean**, but Class B's marks are much more spread out.

This difference is measured using **Measures of Dispersion**.

Importance in Data Science

Measures of Dispersion help us:

- Understand data variability.
- Detect outliers.
- Compare datasets.
- Measure consistency.
- Improve Machine Learning models.
- Identify unusual observations.

- Perform feature engineering.
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Types of Measures of Dispersion

There are five major measures:

1. **Range**
 2. **Interquartile Range (IQR)**
 3. **Mean Absolute Deviation (MAD)**
 4. **Variance**
 5. **Standard Deviation**
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1. Range

The **Range** is the simplest measure of dispersion.

It is the difference between the **largest** and **smallest** value.

FORMULA

$$\text{Range} = \text{Maximum Value} - \text{Minimum Value}$$

EXAMPLE

Dataset

12, 18, 20, 25, 30

Maximum = **30**

Minimum = **12**

Range

$$30 - 12 = 18$$

Answer: Range = 18

ADVANTAGES

- Very easy to calculate.
- Quick understanding of spread.

DISADVANTAGES

- Uses only two values.
- Highly affected by outliers.

EXAMPLE OF OUTLIER

Dataset

```
5, 6, 7, 8, 100
```

Range

```
100 - 5 = 95
```

The value **100** greatly increases the range, even though the other values are close together.

2. Interquartile Range (IQR)

The **Interquartile Range (IQR)** measures the spread of the **middle 50%** of the data.

It is less affected by outliers than the range.

QUARTILES

- **Q1** = First Quartile (25%)
- **Q2** = Median (50%)
- **Q3** = Third Quartile (75%)

FORMULA

```
[  
\text{IQR} = Q_3 - Q_1  
]
```

EXAMPLE

Dataset

```
5, 8, 10, 12, 15, 18, 20, 25, 30
```

- **Q1 = 9**
- **Q3 = 22.5**

IQR

```
22.5 - 9 = 13.5
```

ADVANTAGES

- Not affected much by outliers.
 - Best for skewed data.
 - Useful for detecting outliers.
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DISADVANTAGES

- Ignores the lowest 25% and highest 25% of the data.
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3. Mean Absolute Deviation (MAD)

The **Mean Absolute Deviation** is the average of the **absolute differences** between each observation and the mean.

FORMULA

$$\text{MAD} = \frac{\sum |x_i - \bar{x}|}{n}$$

EXAMPLE

Dataset

2, 4, 6, 8, 10

Mean

6

Absolute deviations

$|2-6| = 4$
 $|4-6| = 2$
 $|6-6| = 0$
 $|8-6| = 2$
 $|10-6| = 4$

Sum

$4+2+0+2+4 = 12$

MAD

$$12/5 = 2.4$$

Answer: MAD = 2.4

ADVANTAGES

- Easy to understand.
- Less influenced by extreme values than variance.

DISADVANTAGES

- Used less frequently in advanced statistics than variance and standard deviation.
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4. Variance

Variance measures the **average squared deviation** of each observation from the mean.

A **small variance** means data points are close to the mean.

A **large variance** means data points are widely spread.

POPULATION VARIANCE FORMULA

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

SAMPLE VARIANCE FORMULA

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

```
genui {"descriptive_statistics_sampling_learning_block_staging":  
{"type_id": "VARIANCE"}}
```

EXAMPLE

Dataset

2, 4, 6

Mean

4

Squared deviations

$$(2-4)^2 = 4$$

$$(4-4)^2 = 0$$

$$(6-4)^2 = 4$$

Variance

$$(4+0+4)/3 = 2.67$$

Answer: Variance ≈ 2.67

ADVANTAGES

- Uses every observation.
- Important in Machine Learning and statistical analysis.

DISADVANTAGES

- Measured in squared units, making interpretation less intuitive.
-

5. Standard Deviation

Standard Deviation is the **square root of the variance**.

It is the most widely used measure of dispersion because it is expressed in the **same units as the original data**.

POPULATION FORMULA

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

SAMPLE FORMULA

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

`genui{"descriptive_statistics_sampling_learning_block_staging":
{"type_id":"STANDARD_DEVIATION"}}`

EXAMPLE

Dataset

2, 4, 6

Variance

2.67

Standard Deviation

$\sqrt{2.67} \approx 1.63$

Answer: Standard Deviation $\approx$ **1.63**

INTERPRETATION

Suppose the average height of students is **170 cm**.

- Standard Deviation = **2 cm** → Heights are very similar.
- Standard Deviation = **15 cm** → Heights vary significantly.

ADVANTAGES

- Uses all observations.
- Same unit as the original data.
- Most commonly used in statistics and machine learning.
- Essential for algorithms such as Linear Regression, Logistic Regression, PCA, and K-Means.

DISADVANTAGES

- Sensitive to outliers.

Variance vs. Standard Deviation

Feature	Variance	Standard Deviation
Unit	Squared units	Original units
Formula	Average squared deviation	Square root of variance

Interpretation	Harder	Easier
Use	Mathematical analysis	Practical interpretation

Comparison of Measures of Dispersion

Measure	Formula	Uses All Data	Affected by Outliers	Best Use
Range	Max – Min	✗	✓	Quick estimate of spread
IQR	Q3 – Q1	✗	✗	Skewed data, outlier detection
MAD	Mean of absolute deviations	✓	Less	General variability
Variance	Average squared deviation	✓	✓	Statistical analysis, ML
Standard Deviation	$\sqrt{\text{Variance}}$	✓	✓	Data Science and ML

Real-Life Applications

FINANCE

Measure the volatility of stock prices.

EDUCATION

Analyze variation in student marks.

MANUFACTURING

Check consistency in product quality.

HEALTHCARE

Measure variation in patient blood pressure.

WEATHER

Study fluctuations in temperature.

DATA SCIENCE

- Feature scaling
 - Outlier detection
 - Exploratory Data Analysis (EDA)
 - Model evaluation
 - Data preprocessing
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Python Example

```
import statistics

data = [12, 15, 18, 20, 25]

print("Range :", max(data) - min(data))
print("Variance :", statistics.variance(data))
print("Standard Deviation :", statistics.stdev(data))
```

OUTPUT

```
Range : 13
Variance : 24.3
Standard Deviation : 4.93
```

Summary

Measure	Meaning	Best Used For
Range	Difference between maximum and minimum values	Quick measure of spread
Interquartile Range (IQR)	Spread of the middle 50% of the data	Skewed data and outlier detection
Mean Absolute Deviation (MAD)	Average absolute distance from the mean	Measuring general variability
Variance	Average squared deviation from the mean	Statistical analysis and machine learning
Standard Deviation	Square root of variance	Understanding overall data spread and comparing datasets

KEY TAKEAWAY

- **Measures of Central Tendency** describe the **center** of the data.
- **Measures of Dispersion** describe the **spread** of the data.
- Among all dispersion measures, **Standard Deviation** is the most widely used in **Data Science** because it provides a meaningful measure of variability in the same units as the original data and is fundamental to many statistical and machine learning techniques.